STAT X12, Lab 6: Midterm Review Problems

Solutions will be posted Monday 3/9

This problem set is intended to give you more practice with core concepts related to multiple linear regression models (including multi-factor studies like TWA) and interpreting output from applying these models to observed data as you prepare for the midterm exam. ***This set of problems does not encompass all the concepts we have covered this semester and should not be thought of as exhausted.***

Other great study resources include

1. **Conceptual problems from your text** (at the ends of each chapter)
2. Quizzes
3. Worksheets from class
4. Activities and Labs
5. Homework assignments

You should be able to take a study prompt, identify the research question, write an appropriate theoretical statistical model for addressing the question (and understand the assumptions that come with the model), know how to visualize raw data, fit the model to observed data and generate useful summary graphics of the model results, assess model assumptions using residual diagnostics (and your brain), and accurately interpret output from statistical software. You should also be able to write a scope of inference specific to the study.

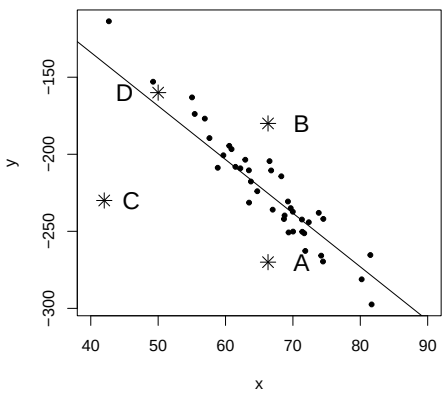
Conceptual Questions [10 pts]

1. [2 pts] One day you are reading a paper that contained the results of an interesting study. The researchers were reporting results of a two-sample t-test with a p-value of 0.675. They stated that "There is no difference in the true mean of the response between the groups." Briefly (one or two sentences) critique this claim.

2. [4 pts] Suppose researchers want to investigate the relationship between weights of cutthroat trout and section of the Bitteroot river in SW Montana. The river was split up into 7 segments. Each segment was electrofished and weights (gm) were collected for all of the fish at each site. The sample sizes were different in each of the 7 segments.

    a. [4 pts] Write a question you would ask the researchers who designed this study that could help you better assess whether the independence assumption is severely violated. Explain why you are asking your question/how it will help you assess this assumptiton.

3. [4 pts] Consider a simple linear regression setting (modeling a response, denoted y, with one cts. covariate, denoted x). Which of the labeled points (A,B,C,D) has the HIGHEST potential for influence on regression results (estimated line uses all points)? Briefly explain why? *Use the scatterplot below.*



Activity level and cholesterol Suppose that you are interested in estimating the difference in total cholesterol levels (milligrams per decaLiter (mg/dL)) in three groups of subjects, after accounting for their age (years). The subjects are characterized as *Active*, *Somewhat Active*, and *Inactive*. Each subject was measured once during an annual well-check event, and the data were collected using a random sample of $n = 150$ subjects from a university health care system. A `glimpse` of the data frame `df_chol` is provided below.

```
## Rows: 150
## Columns: 3
## $ Status <chr> "Active", "Active", "Active", "Active", "Active", "Active", "Ac~
## $ Age    <dbl> 52.44068, 59.40979, 54.24420, 59.87561, 40.54164, 27.48673, 34.~
## $ Chol   <dbl> 166.1454, 144.2535, 138.1953, 138.2277, 196.0881, 214.3865, 207~
```

4. [25 pts] Suppose you naively fit the following model without looking at the data first:

```
##
## Call:
## lm(formula = Chol ~ Status + Age, data = df_chol)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -59.936 -16.181  -1.743  15.273  57.550
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    197.0737     6.7283  29.290  < 2e-16
## StatusInactive  55.9092     5.9497   9.397  < 2e-16
## StatusSomewhat  20.3713     4.6696   4.363 2.42e-05
## Age             -0.4362     0.1387  -3.145  0.00201
##
```

```
## Residual standard error: 23.33 on 146 degrees of freedom
## Multiple R-squared:  0.4074, Adjusted R-squared:  0.3952
## F-statistic: 33.45 on 3 and 146 DF,  p-value: < 2.2e-16
```

```
##                      2.5 %      97.5 %
## (Intercept)    183.7761286 210.3711919
## StatusInactive  44.1504898  67.6679157
## StatusSomewhat  11.1425615  29.6000091
## Age             -0.7102719  -0.1621317
```

a. [10 pts] Write out the complete assumed population model that was fit to these data, defining all variables.

b. [5 pts] Draw a sketch of the assumed population model that was fit to these data and explain, in words, the assumption the model makes about the relationship between age and cholesterol across the activity levels.

c. [5 pts] Write a summary of statistical findings for the partial regression coefficient associated with `Age`, including a 95% CI for the parameter.

d. [5 pts] When you show your results to the your advisor, they ask you about the potential for an interaction between Age and Activity Status on Cholesterol levels. Describe *in words* what an interaction between Age and Activity Status would mean in the context of this problem.

5. [6 pts] You return to your computer and fit the interaction model and compare it to the model you fit previously.

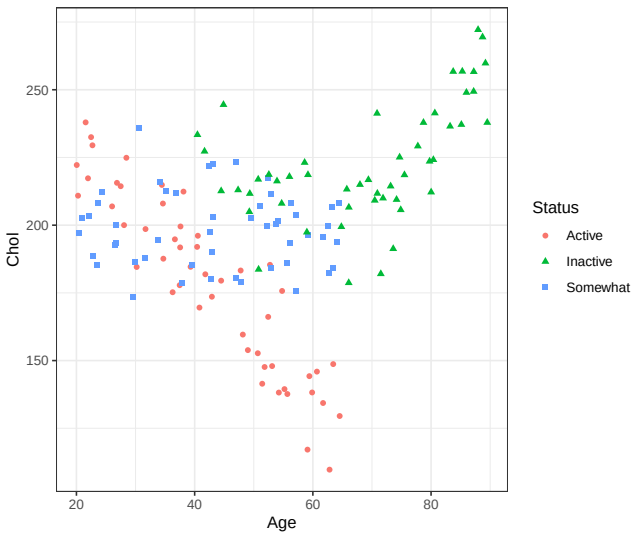
```
## Analysis of Variance Table
##
## Model 1: Chol ~ Status + Age
## Model 2: Chol ~ Status * Age
##   Res.Df   RSS Df Sum of Sq      F    Pr(>F)
## 1    146 79474
## 2    144 34661  2     44813 93.089 < 2.2e-16
```

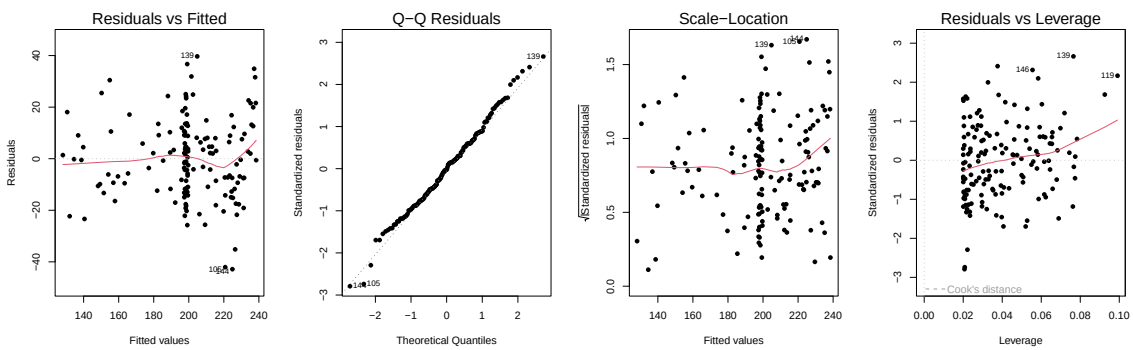
a. [4 pts] Use the output to write a summary of statistical findings for the comparison.

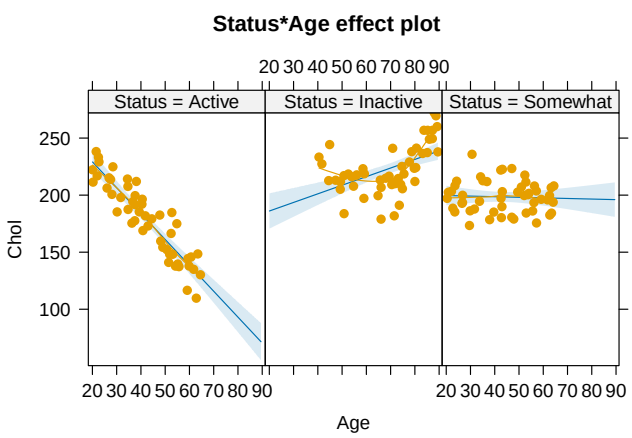
b. [2 pts] Using the interaction model output below, compute the estimated mean Cholesterol level for a subject who is 30 years old and inactive. Show your work.

```
##       Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
## -103.56372  -55.95349   -0.02588   16.68420    2.82088  274.54693
```

5. [5 pts] After you've fit a model to address your advisor's question, it hits you…you didn't check any of the diagnostics for either MLR, or plot the data *whoops!* Raw data plots and diagnostic plots are provided for the interaction model. Use the diagnostics to explain whether you are okay proceeding with the interaction model. If you are not comfortable proceeding, be specific about your concerns. *Explain your reasoning clearly!*







- **BONUS [up to 2 pts]:** Propose a next step in the model refinement/expansion process based off your assessment of diagnostic plots.

6. [55 pts] You bring your summary of the interaction results back to your advisor and they suggest that simplifying the `Age` variable used above will make sense for this investigation. Specifically, they suggest creating a categorical variable from it with the following code.

a. [10 pts] Write out a theoretical model that investigates the relationship between average cholesterol, level of activity, and the categorical version of age; considering the potential

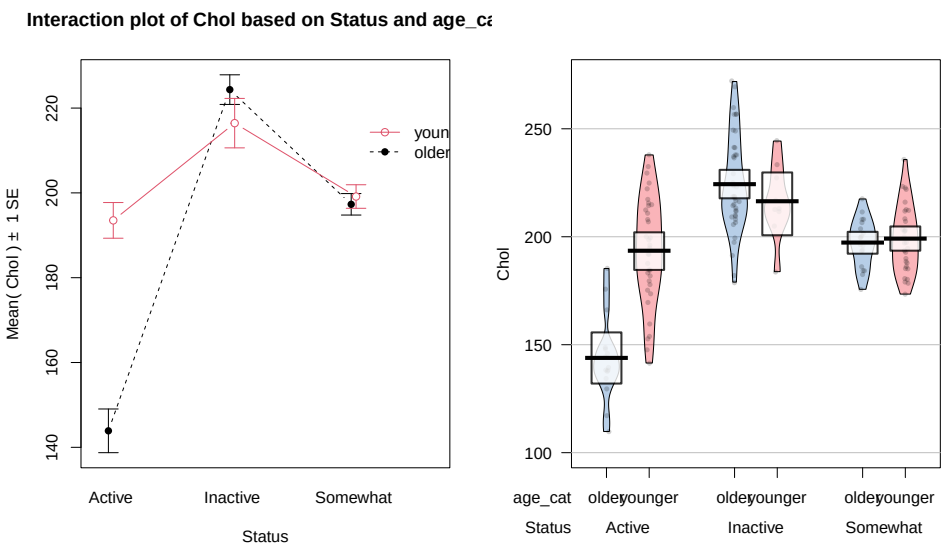
for the relationship between activity and average cholesterol to differ across the two age groups. *Define all variables and identify which parameters allow you to address the question of interest.*

b. [2 pts] Summary statistics for addressing the research question are provided for these data, is this a balanced design? Explain.

```
## `summarise()` has grouped output by 'age_cat'. You can override using the
## `.groups` argument.
```

```
## # A tibble: 6 x 4
## # Groups:   age_cat [2]
##   age_cat Status   avg_chol     n
##   <chr>   <chr>       <dbl> <int>
## 1 older   Active       144.    15
## 2 older   Inactive     224.    41
## 3 older   Somewhat     197.    19
## 4 younger Active       194.    35
## 5 younger Inactive     216.     9
## 6 younger Somewhat     199.    31
```

c. [5 pts] A raw data visualization and an interaction plot are provided for these data. Does there appear to be an interaction between `Status` and `age_cat` as they relate to cholesterol? Identify relevant features in an appropriate plot.



d. [2 pts] An interaction model is fit to these data and a two-way ANOVA table is provided. Write the two lines of code that generated this output.

```
## Analysis of Variance Table
##
## Response: Chol
##                 Df Sum Sq Mean Sq F value    Pr(>F)
## age_cat          1    297   296.9  0.7308    0.3941
```

```
## Status           2  56854 28426.9 69.9756 < 2.2e-16
## age_cat:Status   2  18451  9225.4 22.7093 2.677e-09
## Residuals      144  58499   406.2
```

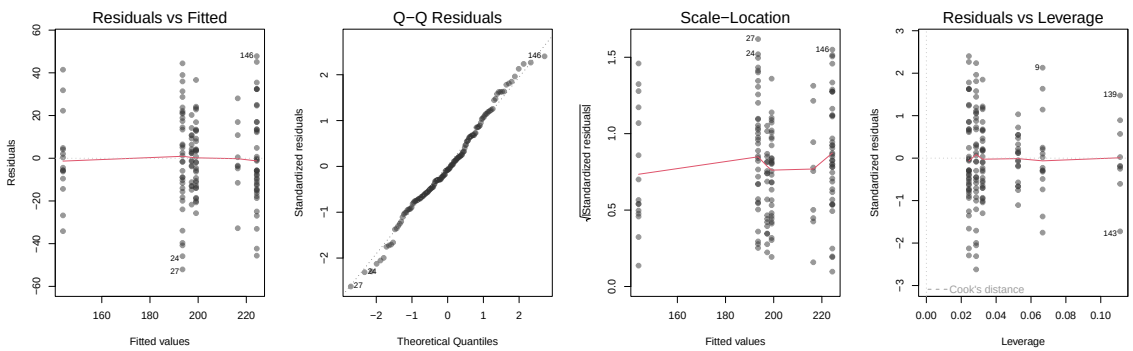
e. [6 pts] What are the null and alternative hypotheses being tested in the bottom row of the anova table? Please define each hypothesis in mathematical notation (i.e., in terms of βs in the linear model) and in words.

f. [5 pts] Write a summary of statistical findings for the hypothesis test described in (f), *be sure to include the distribution the test statistic follows under the null hypothesis.*

g. [10 pts] Make a sketch of an appropriate type of plot that you could use to help communicate what you learned (in practical terms) from this analysis.

h. [5 pts] Write a scope of inference for this study.

i. [5 pts] Refer to the diagnostic plots for this model, do you trust the inferences you made above? Briefly explain.



k. [5 pts] Set up a contrast that would allow you to further investigate how different the effect of age group is for Active subjects compared to Somewhat Active subjects in the study. The levels of the interaction between `age_cat` and `Status` are provided below.

```
## [1] "older.Active"     "younger.Active"   "older.Inactive"   "younger.Inactive"
## [5] "older.Somewhat"   "younger.Somewhat"
```